

# Tarea1-DanielGonzalez

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## 1 Tarea 1 DAAML

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- 1.0.1 1. Cargar la base de datos en el ambiente. Identifique los tipos de datos que se encuentran en la base, realice estadísticas descriptivas sobre las variables importantes (Hint: Revisar la distribuciones, datos faltantes, outliers, etc.) y limpie las variables cuando sea necesario. Genere una variable binaria para indicar quienes dieron el test. Justifique su proceso.

```
[1]: import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import statsmodels.api as sm
import statsmodels.formula.api as smf
import sklearn
import scipy
from scipy.stats import nbinom
import seaborn as sns
from statsmodels.iolib.summary2 import summary_col
import missingno as msno
import warnings
warnings.filterwarnings("ignore")
from stargazer.stargazer import Stargazer
from IPython.core.display import HTML

%matplotlib inline
```

```
[2]: # importamos los datos
df = pd.read_csv("./data/student_productivity.csv", decimal='.')
# creamos una copia de los datos
data = df.copy()
```

```
[3]: data.head()
```

```
[3]:   student_id  age  gender  academic_level  study_hours  self_study_hours  \
0           1  20.0   NaN    Undergraduate         5.37             2.09
1           2  16.0 Female      High School         5.85             5.04
```

|   |   |      |        |               |      |      |
|---|---|------|--------|---------------|------|------|
| 2 | 3 | 18.0 | Female | Undergraduate | 5.69 | 2.27 |
| 3 | 4 | 24.0 | Male   | Undergraduate | 2.32 | 1.06 |
| 4 | 5 | 24.0 | Female | Postgraduate  | 3.87 | 2.63 |

|   | online_classes_hours | social_media_hours | gaming_hours | sleep_hours | ... | \ |
|---|----------------------|--------------------|--------------|-------------|-----|---|
| 0 | 1.85                 | 3.66               | 2.32         | 7.73        | ... |   |
| 1 | 1.87                 | 3.60               | 2.79         | 6.11        | ... |   |
| 2 | 0.00                 | 2.93               | 3.94         | 7.11        | ... |   |
| 3 | 2.30                 | 4.34               | 2.37         | 8.54        | ... |   |
| 4 | 2.54                 | 3.71               | 1.30         | 7.69        | ... |   |

|   | caffeine_intake_mg | part_time_job | upcoming_deadline | internet_quality | \ |
|---|--------------------|---------------|-------------------|------------------|---|
| 0 | 475.0              | No            | 0.0               | Good             |   |
| 1 | 362.0              | no            | 0.0               | Good             |   |
| 2 | 200.0              | No            | 0.0               | Average          |   |
| 3 | 233.0              | No            | 1.0               | Poor             |   |
| 4 | 159.0              | No            | 1.0               | Poor             |   |

|   | mental_health_score | drug_use | focus_index | burnout_level | \ |
|---|---------------------|----------|-------------|---------------|---|
| 0 | 3.0                 | NaN      | 19.01       | 31.77         |   |
| 1 | 10.0                | NaN      | 42.10       | 45.89         |   |
| 2 | 5.0                 | NaN      | 21.93       | 37.07         |   |
| 3 | 3.0                 | 1.0      | 13.47       | 43.63         |   |
| 4 | 2.0                 | NaN      | 19.95       | 56.62         |   |

|   | productivity_score | exam_score |
|---|--------------------|------------|
| 0 | 42.59              | 25.08      |
| 1 | 67.15              | 37.83      |
| 2 | 37.68              | 18.66      |
| 3 | 12.83              | 1.00       |
| 4 | 18.53              | 7.78       |

[5 rows x 22 columns]

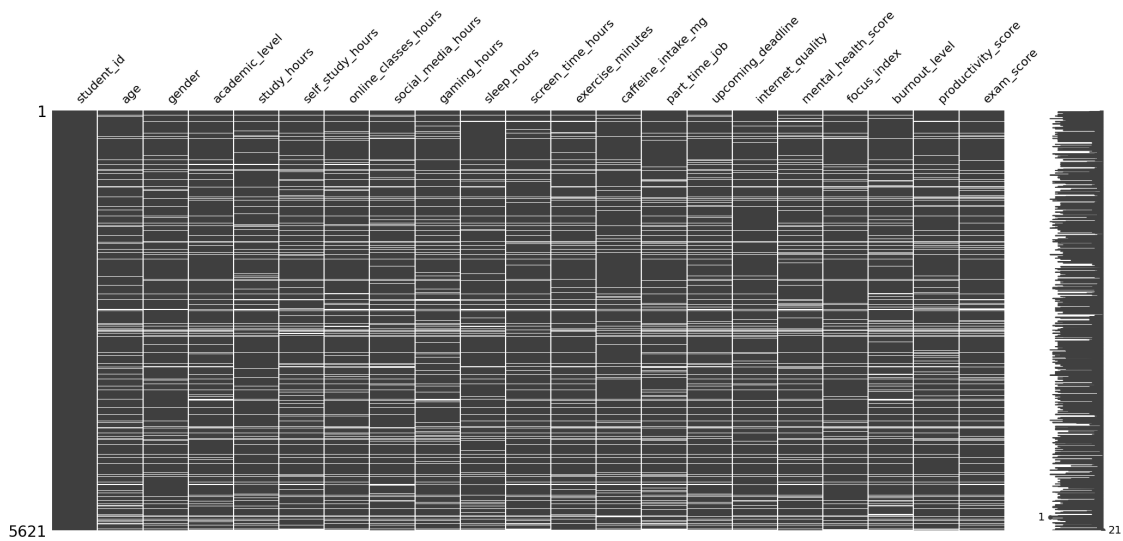
Según la información de Data Wrangler, la variable “drug\_use” contiene un 73% de datos faltantes, por lo que es mejor eliminarla.

```
[4]: data.drop('drug_use', axis=1, inplace=True)
```

Verificamos los datos faltantes:

```
[5]: msno.matrix(data)
```

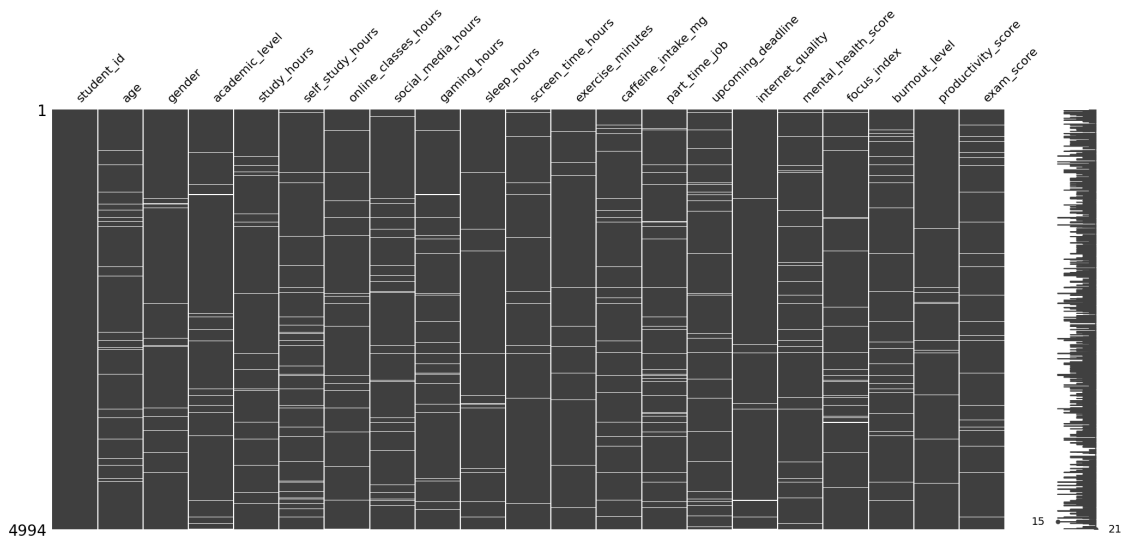
```
[5]: <Axes: >
```



Se puede apreciar una cantidad significativamente grande de datos faltantes, pero no existen filas vacías, por lo que es necesario hacer un filtro y eliminar filas con cierta cantidad de datos vacíos

```
[6]: # Eliminamos las filas con más de 5 valores faltantes
data = data.dropna(thresh=data.shape[1] - 6)
msno.matrix(data)
```

[6]: <Axes: >



Transformamos datos

```
[7]: # La variable 'exercise_minutes' es del tipo string, por lo que la convertimos
      ↪ a numérica
data['exercise_minutes'] = pd.to_numeric(data['exercise_minutes'],
      ↪ errors='coerce')
# La variable 'part_time_job' tiene datos 'No', 'no', 'Yes' y 'yes' que pueden
      ↪ ser reemplazados por 0 y 1
data['part_time_job'] = data['part_time_job'].replace({'No': 0, 'no': 0, 'Yes':
      ↪ 1, 'yes': 1})
# La variable 'gender' tiene datos 'Male' y 'Female' que pueden ser
      ↪ reemplazados por 0 y 1 (también tiene 'other' pero es un caso muy raro, por
      ↪ lo que lo dejaremos como NA)
data['gender'] = data['gender'].replace({'Male': 0, 'Female': 1})
# Para dejar 'Other' como NA, podemos usar el método replace para convertirlo a
      ↪ NaN
data['gender'] = data['gender'].replace({'Other': np.nan})
data.head()
```

```
[7]:
```

|   | student_id | age  | gender | academic_level | study_hours | self_study_hours | \ |
|---|------------|------|--------|----------------|-------------|------------------|---|
| 0 | 1          | 20.0 | NaN    | Undergraduate  | 5.37        | 2.09             |   |
| 1 | 2          | 16.0 | 1.0    | High School    | 5.85        | 5.04             |   |
| 2 | 3          | 18.0 | 1.0    | Undergraduate  | 5.69        | 2.27             |   |
| 3 | 4          | 24.0 | 0.0    | Undergraduate  | 2.32        | 1.06             |   |
| 4 | 5          | 24.0 | 1.0    | Postgraduate   | 3.87        | 2.63             |   |

|   | online_classes_hours | social_media_hours | gaming_hours | sleep_hours | ... | \ |
|---|----------------------|--------------------|--------------|-------------|-----|---|
| 0 | 1.85                 | 3.66               | 2.32         | 7.73        | ... |   |
| 1 | 1.87                 | 3.60               | 2.79         | 6.11        | ... |   |
| 2 | 0.00                 | 2.93               | 3.94         | 7.11        | ... |   |
| 3 | 2.30                 | 4.34               | 2.37         | 8.54        | ... |   |
| 4 | 2.54                 | 3.71               | 1.30         | 7.69        | ... |   |

|   | exercise_minutes | caffeine_intake_mg | part_time_job | upcoming_deadline | \ |
|---|------------------|--------------------|---------------|-------------------|---|
| 0 | 54.0             | 475.0              | 0.0           | 0.0               |   |
| 1 | 63.0             | 362.0              | 0.0           | 0.0               |   |
| 2 | 28.0             | 200.0              | 0.0           | 0.0               |   |
| 3 | 55.0             | 233.0              | 0.0           | 1.0               |   |
| 4 | 60.0             | 159.0              | 0.0           | 1.0               |   |

|   | internet_quality | mental_health_score | focus_index | burnout_level | \ |
|---|------------------|---------------------|-------------|---------------|---|
| 0 | Good             | 3.0                 | 19.01       | 31.77         |   |
| 1 | Good             | 10.0                | 42.10       | 45.89         |   |
| 2 | Average          | 5.0                 | 21.93       | 37.07         |   |
| 3 | Poor             | 3.0                 | 13.47       | 43.63         |   |
| 4 | Poor             | 2.0                 | 19.95       | 56.62         |   |

|   | productivity_score | exam_score |
|---|--------------------|------------|
| 0 | 42.59              | 25.08      |

|   |       |       |
|---|-------|-------|
| 1 | 67.15 | 37.83 |
| 2 | 37.68 | 18.66 |
| 3 | 12.83 | 1.00  |
| 4 | 18.53 | 7.78  |

[5 rows x 21 columns]

Hay que completar los datos faltantes con la media, mediana o moda según corresponda. Tomando en cuenta que, viendo los histogramas, la mayoría de datos distribuyen normal, salvo algunos que tienen ciertas asimetrías, podemos rellenar los datos faltantes de estas variables con la media.

```
[8]: # Completamos los datos faltantes con la media (redondeando a 2 decimales) para
      ↳ las columnas age, study_hours, self_study_hours, online_classes_hours,
      ↳ social_media_hours, sleep_hours, screen_time_hours, exercise_minutes,
      ↳ caffeine_intake_mg, mental_health_score, focus_index, burnout_level,
      ↳ productivity_score
data['age'] = data['age'].fillna(round(data['age'].mean(), 2))
data['study_hours'] = data['study_hours'].fillna(round(data['study_hours'].
      ↳ mean(), 2))
data['self_study_hours'] = data['self_study_hours'].
      ↳ fillna(round(data['self_study_hours'].mean(), 2))
data['online_classes_hours'] = data['online_classes_hours'].
      ↳ fillna(round(data['online_classes_hours'].mean(), 2))
data['social_media_hours'] = data['social_media_hours'].
      ↳ fillna(round(data['social_media_hours'].mean(), 2))
data['sleep_hours'] = data['sleep_hours'].fillna(round(data['sleep_hours'].
      ↳ mean(), 2))
data['screen_time_hours'] = data['screen_time_hours'].
      ↳ fillna(round(data['screen_time_hours'].mean(), 2))
data['exercise_minutes'] = data['exercise_minutes'].
      ↳ fillna(round(data['exercise_minutes'].mean(), 2))
data['caffeine_intake_mg'] = data['caffeine_intake_mg'].
      ↳ fillna(round(data['caffeine_intake_mg'].mean(), 2))
data['mental_health_score'] = data['mental_health_score'].
      ↳ fillna(round(data['mental_health_score'].mean(), 2))
data['focus_index'] = data['focus_index'].fillna(round(data['focus_index'].
      ↳ mean(), 2))
data['burnout_level'] = data['burnout_level'].
      ↳ fillna(round(data['burnout_level'].mean(), 2))
data['productivity_score'] = data['productivity_score'].
      ↳ fillna(round(data['productivity_score'].mean(), 2))
```

Para las variables binarias, se utiliza la moda

```
[9]: # Para las variables binarias part_time_job, upcoming_deadline y gender,
      ↳ completamos los datos faltantes con la moda
```

```
data['part_time_job'] = data['part_time_job'].fillna(data['part_time_job'].
    ↪mode()[0])
data['upcoming_deadline'] = data['upcoming_deadline'].
    ↪fillna(data['upcoming_deadline'].mode()[0])
data['gender'] = data['gender'].fillna(data['gender'].mode()[0])
data.head()
```

```
[9]:
```

|   | student_id | age  | gender | academic_level | study_hours | self_study_hours | \ |
|---|------------|------|--------|----------------|-------------|------------------|---|
| 0 | 1          | 20.0 | 0.0    | Undergraduate  | 5.37        | 2.09             |   |
| 1 | 2          | 16.0 | 1.0    | High School    | 5.85        | 5.04             |   |
| 2 | 3          | 18.0 | 1.0    | Undergraduate  | 5.69        | 2.27             |   |
| 3 | 4          | 24.0 | 0.0    | Undergraduate  | 2.32        | 1.06             |   |
| 4 | 5          | 24.0 | 1.0    | Postgraduate   | 3.87        | 2.63             |   |

|   | online_classes_hours | social_media_hours | gaming_hours | sleep_hours | ... | \ |
|---|----------------------|--------------------|--------------|-------------|-----|---|
| 0 | 1.85                 | 3.66               | 2.32         | 7.73        | ... |   |
| 1 | 1.87                 | 3.60               | 2.79         | 6.11        | ... |   |
| 2 | 0.00                 | 2.93               | 3.94         | 7.11        | ... |   |
| 3 | 2.30                 | 4.34               | 2.37         | 8.54        | ... |   |
| 4 | 2.54                 | 3.71               | 1.30         | 7.69        | ... |   |

|   | exercise_minutes | caffeine_intake_mg | part_time_job | upcoming_deadline | \ |
|---|------------------|--------------------|---------------|-------------------|---|
| 0 | 54.0             | 475.0              | 0.0           | 0.0               |   |
| 1 | 63.0             | 362.0              | 0.0           | 0.0               |   |
| 2 | 28.0             | 200.0              | 0.0           | 0.0               |   |
| 3 | 55.0             | 233.0              | 0.0           | 1.0               |   |
| 4 | 60.0             | 159.0              | 0.0           | 1.0               |   |

|   | internet_quality | mental_health_score | focus_index | burnout_level | \ |
|---|------------------|---------------------|-------------|---------------|---|
| 0 | Good             | 3.0                 | 19.01       | 31.77         |   |
| 1 | Good             | 10.0                | 42.10       | 45.89         |   |
| 2 | Average          | 5.0                 | 21.93       | 37.07         |   |
| 3 | Poor             | 3.0                 | 13.47       | 43.63         |   |
| 4 | Poor             | 2.0                 | 19.95       | 56.62         |   |

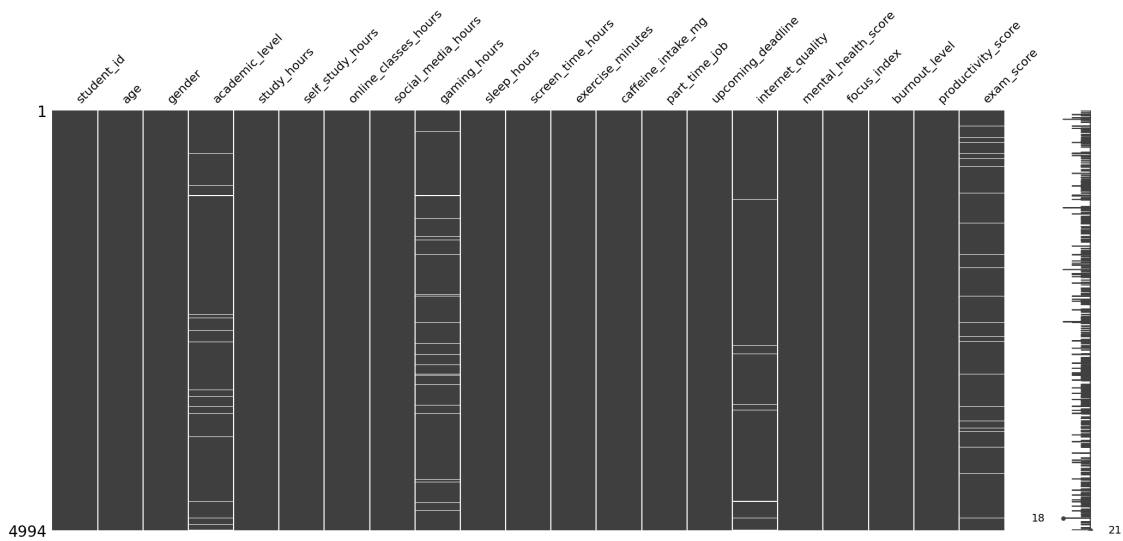
  

|   | productivity_score | exam_score |
|---|--------------------|------------|
| 0 | 42.59              | 25.08      |
| 1 | 67.15              | 37.83      |
| 2 | 37.68              | 18.66      |
| 3 | 12.83              | 1.00       |
| 4 | 18.53              | 7.78       |

[5 rows x 21 columns]

```
[10]: msno.matrix(data)
```

```
[10]: <Axes: >
```



Creamos una nueva variable para saber si es que el alumno rindió o no el examen

```
[11]: # Generamos una nueva columna llamada 'rinde' que es 0 cuando 'exam_score' es 1
      ↪ y 1 cuando 'exam_score' es cualquier otro valor
data['rinde'] = data['exam_score'].apply(lambda x: 0 if x == 1 else 1)
data.head()
```

```
[11]:  student_id  age  gender  academic_level  study_hours  self_study_hours  \
0          1  20.0    0.0    Undergraduate        5.37          2.09
1          2  16.0    1.0      High School        5.85          5.04
2          3  18.0    1.0    Undergraduate        5.69          2.27
3          4  24.0    0.0    Undergraduate        2.32          1.06
4          5  24.0    1.0    Postgraduate        3.87          2.63

      online_classes_hours  social_media_hours  gaming_hours  sleep_hours  ...  \
0                1.85                3.66          2.32          7.73  ...
1                1.87                3.60          2.79          6.11  ...
2                0.00                2.93          3.94          7.11  ...
3                2.30                4.34          2.37          8.54  ...
4                2.54                3.71          1.30          7.69  ...

      caffeine_intake_mg  part_time_job  upcoming_deadline  internet_quality  \
0                475.0          0.0          0.0          Good
1                362.0          0.0          0.0          Good
2                200.0          0.0          0.0        Average
3                233.0          0.0          1.0          Poor
4                159.0          0.0          1.0          Poor

      mental_health_score  focus_index  burnout_level  productivity_score  \
```

|   |      |       |       |       |
|---|------|-------|-------|-------|
| 0 | 3.0  | 19.01 | 31.77 | 42.59 |
| 1 | 10.0 | 42.10 | 45.89 | 67.15 |
| 2 | 5.0  | 21.93 | 37.07 | 37.68 |
| 3 | 3.0  | 13.47 | 43.63 | 12.83 |
| 4 | 2.0  | 19.95 | 56.62 | 18.53 |

|   | exam_score | rinde |
|---|------------|-------|
| 0 | 25.08      | 1     |
| 1 | 37.83      | 1     |
| 2 | 18.66      | 1     |
| 3 | 1.00       | 0     |
| 4 | 7.78       | 1     |

[5 rows x 22 columns]

Finalmente seguimos llenando los datos NaN con respecto a los criterios vistos anteriormente

```
[12]: # Ahora, completamos los datos faltantes de 'exam_score' con la media, pero
      ↪ solo para los casos donde 'rinde' es 1 (es decir, donde 'exam_score' no es 1)
data.loc[data['rinde'] == 1, 'exam_score'] = data.loc[data['rinde'] == 1,
      ↪ 'exam_score'].fillna(data.loc[data['rinde'] == 1, 'exam_score'].mean())
msno.matrix(data)
data
```

```
[12]:
```

|      | student_id | age  | gender | academic_level | study_hours | self_study_hours | \ |
|------|------------|------|--------|----------------|-------------|------------------|---|
| 0    | 1          | 20.0 | 0.0    | Undergraduate  | 5.37        | 2.09             |   |
| 1    | 2          | 16.0 | 1.0    | High School    | 5.85        | 5.04             |   |
| 2    | 3          | 18.0 | 1.0    | Undergraduate  | 5.69        | 2.27             |   |
| 3    | 4          | 24.0 | 0.0    | Undergraduate  | 2.32        | 1.06             |   |
| 4    | 5          | 24.0 | 1.0    | Postgraduate   | 3.87        | 2.63             |   |
| ...  | ...        | ...  | ...    | ...            | ...         | ...              |   |
| 5612 | 5613       | 18.0 | 0.0    | Postgraduate   | 4.30        | 2.34             |   |
| 5614 | 5615       | 25.0 | 1.0    | NaN            | 3.85        | 2.63             |   |
| 5615 | 5616       | 17.0 | 0.0    | High School    | 6.55        | 2.48             |   |
| 5616 | 5617       | 21.0 | 0.0    | Postgraduate   | 4.16        | 0.00             |   |
| 5620 | 5621       | 18.0 | 0.0    | Postgraduate   | 3.96        | 0.98             |   |

|      | online_classes_hours | social_media_hours | gaming_hours | sleep_hours | \ |
|------|----------------------|--------------------|--------------|-------------|---|
| 0    | 1.85                 | 3.66               | 2.32         | 7.73        |   |
| 1    | 1.87                 | 3.60               | 2.79         | 6.11        |   |
| 2    | 0.00                 | 2.93               | 3.94         | 7.11        |   |
| 3    | 2.30                 | 4.34               | 2.37         | 8.54        |   |
| 4    | 2.54                 | 3.71               | 1.30         | 7.69        |   |
| ...  | ...                  | ...                | ...          | ...         |   |
| 5612 | 0.70                 | 0.43               | 4.04         | 7.39        |   |
| 5614 | 2.01                 | 2.55               | 2.46         | 8.15        |   |
| 5615 | 2.01                 | 3.42               | 2.44         | 6.17        |   |



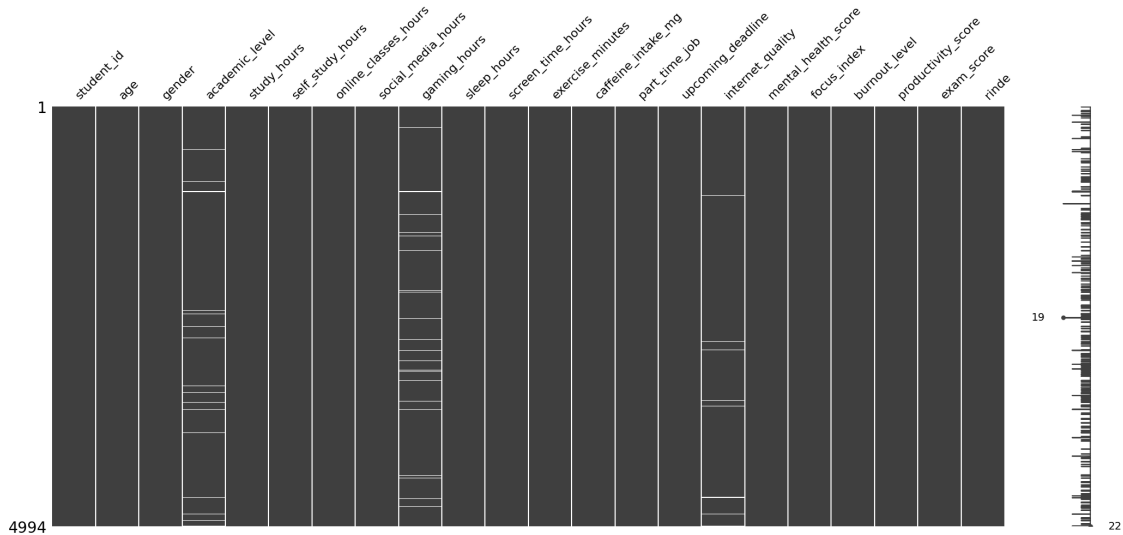
|      |      |      |      |      |
|------|------|------|------|------|
| 5616 | 1.27 | 4.01 | 3.41 | 8.07 |
| 5620 | 1.77 | 4.35 | 2.22 | 6.57 |

|      | ... | caffeine_intake_mg | part_time_job | upcoming_deadline | \ |
|------|-----|--------------------|---------------|-------------------|---|
| 0    | ... | 475.0              | 0.0           | 0.0               |   |
| 1    | ... | 362.0              | 0.0           | 0.0               |   |
| 2    | ... | 200.0              | 0.0           | 0.0               |   |
| 3    | ... | 233.0              | 0.0           | 1.0               |   |
| 4    | ... | 159.0              | 0.0           | 1.0               |   |
| ...  | ... | ...                | ...           | ...               |   |
| 5612 | ... | 87.0               | 0.0           | 1.0               |   |
| 5614 | ... | 453.0              | 1.0           | 1.0               |   |
| 5615 | ... | 173.0              | 1.0           | 1.0               |   |
| 5616 | ... | 152.0              | 0.0           | 1.0               |   |
| 5620 | ... | 154.0              | 0.0           | 0.0               |   |

|      | internet_quality | mental_health_score | focus_index | burnout_level | \ |
|------|------------------|---------------------|-------------|---------------|---|
| 0    | Good             | 3.0                 | 19.01       | 31.77         |   |
| 1    | Good             | 10.0                | 42.10       | 45.89         |   |
| 2    | Average          | 5.0                 | 21.93       | 37.07         |   |
| 3    | Poor             | 3.0                 | 13.47       | 43.63         |   |
| 4    | Poor             | 2.0                 | 19.95       | 56.62         |   |
| ...  | ...              | ...                 | ...         | ...           |   |
| 5612 | Poor             | 2.0                 | 36.18       | 40.42         |   |
| 5614 | NaN              | 10.0                | 38.26       | 57.39         |   |
| 5615 | Poor             | 7.0                 | 34.70       | 56.14         |   |
| 5616 | Poor             | 7.0                 | 21.78       | 48.07         |   |
| 5620 | Poor             | 3.0                 | 16.28       | 36.62         |   |

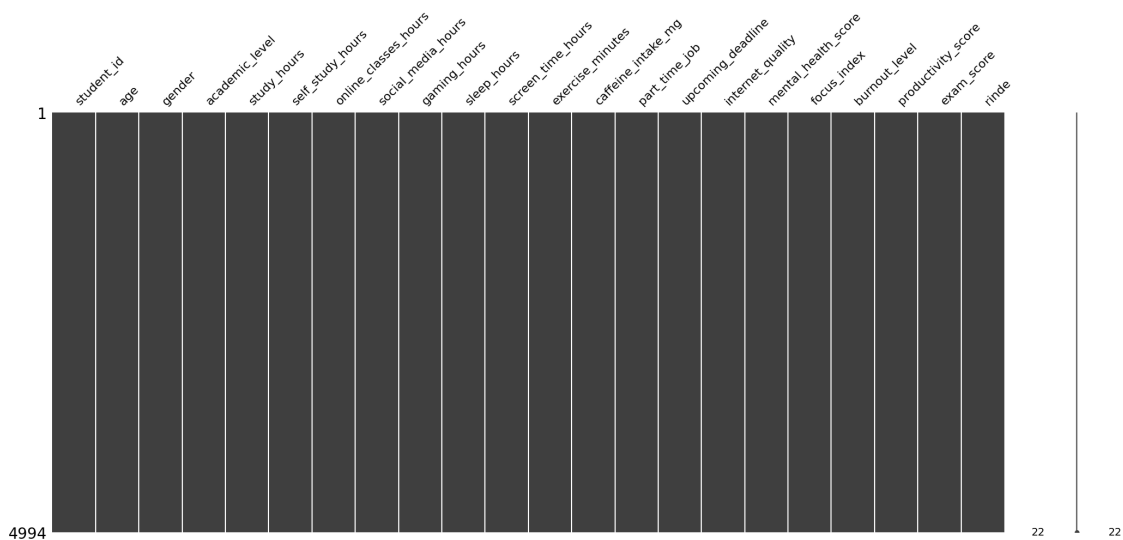
|      | productivity_score | exam_score | rinde |
|------|--------------------|------------|-------|
| 0    | 42.59              | 25.080000  | 1     |
| 1    | 67.15              | 37.830000  | 1     |
| 2    | 37.68              | 18.660000  | 1     |
| 3    | 12.83              | 1.000000   | 0     |
| 4    | 18.53              | 7.780000   | 1     |
| ...  | ...                | ...        | ...   |
| 5612 | 31.42              | 10.570000  | 1     |
| 5614 | 45.30              | 23.770000  | 1     |
| 5615 | 47.66              | 20.616782  | 1     |
| 5616 | 43.06              | 17.150000  | 1     |
| 5620 | 21.36              | 2.810000   | 1     |

[4994 rows x 22 columns]



```
[13]: # para 'academic_level' e 'internet_quality', completamos los datos faltantes
      ↪ con la moda
data['academic_level'] = data['academic_level'].fillna(data['academic_level'].
      ↪ mode()[0])
data['internet_quality'] = data['internet_quality'].
      ↪ fillna(data['internet_quality'].mode()[0])
# para gaming_hours, completamos los datos faltantes con la mediana (datos
      ↪ asimetricos )
data['gaming_hours'] = data['gaming_hours'].fillna(data['gaming_hours'].
      ↪ median())
msno.matrix(data)
```

[13]: <Axes: >



```
[14]: # Transformamos 'academic_level' e 'internet_quality' a variables numericas
      ↪categoricas ordinales
data['academic_level'] = data['academic_level'].replace({'High School': 0,
      ↪'High School ': 0, 'Undergraduate': 1, 'Undergraduate ': 1, 'Postgraduate':
      ↪2, 'Postgraduate ' : 2})
data['internet_quality'] = data['internet_quality'].replace({'Poor': 0,
      ↪'Average': 1, 'Good': 2})
data
```

```
[14]:
```

|      | student_id | age  | gender | academic_level | study_hours | self_study_hours | \ |
|------|------------|------|--------|----------------|-------------|------------------|---|
| 0    | 1          | 20.0 | 0.0    | 1              | 5.37        | 2.09             |   |
| 1    | 2          | 16.0 | 1.0    | 0              | 5.85        | 5.04             |   |
| 2    | 3          | 18.0 | 1.0    | 1              | 5.69        | 2.27             |   |
| 3    | 4          | 24.0 | 0.0    | 1              | 2.32        | 1.06             |   |
| 4    | 5          | 24.0 | 1.0    | 2              | 3.87        | 2.63             |   |
| ...  | ...        | ...  |        | ...            | ...         | ...              |   |
| 5612 | 5613       | 18.0 | 0.0    | 2              | 4.30        | 2.34             |   |
| 5614 | 5615       | 25.0 | 1.0    | 2              | 3.85        | 2.63             |   |
| 5615 | 5616       | 17.0 | 0.0    | 0              | 6.55        | 2.48             |   |
| 5616 | 5617       | 21.0 | 0.0    | 2              | 4.16        | 0.00             |   |
| 5620 | 5621       | 18.0 | 0.0    | 2              | 3.96        | 0.98             |   |

|      | online_classes_hours | social_media_hours | gaming_hours | sleep_hours | \ |
|------|----------------------|--------------------|--------------|-------------|---|
| 0    | 1.85                 | 3.66               | 2.32         | 7.73        |   |
| 1    | 1.87                 | 3.60               | 2.79         | 6.11        |   |
| 2    | 0.00                 | 2.93               | 3.94         | 7.11        |   |
| 3    | 2.30                 | 4.34               | 2.37         | 8.54        |   |
| 4    | 2.54                 | 3.71               | 1.30         | 7.69        |   |
| ...  | ...                  | ...                | ...          | ...         |   |
| 5612 | 0.70                 | 0.43               | 4.04         | 7.39        |   |
| 5614 | 2.01                 | 2.55               | 2.46         | 8.15        |   |
| 5615 | 2.01                 | 3.42               | 2.44         | 6.17        |   |
| 5616 | 1.27                 | 4.01               | 3.41         | 8.07        |   |
| 5620 | 1.77                 | 4.35               | 2.22         | 6.57        |   |

|      | ... | caffeine_intake_mg | part_time_job | upcoming_deadline | \ |
|------|-----|--------------------|---------------|-------------------|---|
| 0    | ... | 475.0              | 0.0           | 0.0               |   |
| 1    | ... | 362.0              | 0.0           | 0.0               |   |
| 2    | ... | 200.0              | 0.0           | 0.0               |   |
| 3    | ... | 233.0              | 0.0           | 1.0               |   |
| 4    | ... | 159.0              | 0.0           | 1.0               |   |
| ...  | ... | ...                | ...           | ...               |   |
| 5612 | ... | 87.0               | 0.0           | 1.0               |   |
| 5614 | ... | 453.0              | 1.0           | 1.0               |   |

|      |     |       |     |     |
|------|-----|-------|-----|-----|
| 5615 | ... | 173.0 | 1.0 | 1.0 |
| 5616 | ... | 152.0 | 0.0 | 1.0 |
| 5620 | ... | 154.0 | 0.0 | 0.0 |

|      | internet_quality | mental_health_score | focus_index | burnout_level | \ |
|------|------------------|---------------------|-------------|---------------|---|
| 0    | 2                | 3.0                 | 19.01       | 31.77         |   |
| 1    | 2                | 10.0                | 42.10       | 45.89         |   |
| 2    | 1                | 5.0                 | 21.93       | 37.07         |   |
| 3    | 0                | 3.0                 | 13.47       | 43.63         |   |
| 4    | 0                | 2.0                 | 19.95       | 56.62         |   |
| ...  | ...              | ...                 | ...         | ...           |   |
| 5612 | 0                | 2.0                 | 36.18       | 40.42         |   |
| 5614 | 2                | 10.0                | 38.26       | 57.39         |   |
| 5615 | 0                | 7.0                 | 34.70       | 56.14         |   |
| 5616 | 0                | 7.0                 | 21.78       | 48.07         |   |
| 5620 | 0                | 3.0                 | 16.28       | 36.62         |   |

|      | productivity_score | exam_score | rinde |
|------|--------------------|------------|-------|
| 0    | 42.59              | 25.080000  | 1     |
| 1    | 67.15              | 37.830000  | 1     |
| 2    | 37.68              | 18.660000  | 1     |
| 3    | 12.83              | 1.000000   | 0     |
| 4    | 18.53              | 7.780000   | 1     |
| ...  | ...                | ...        | ...   |
| 5612 | 31.42              | 10.570000  | 1     |
| 5614 | 45.30              | 23.770000  | 1     |
| 5615 | 47.66              | 20.616782  | 1     |
| 5616 | 43.06              | 17.150000  | 1     |
| 5620 | 21.36              | 2.810000   | 1     |

[4994 rows x 22 columns]

```
[15]: # Creamos variables para si una persona hace ejercicio o no (exercise_minutes > 0), si es que juega videojuegos o no (gaming_hours > 0) y si es que usa redes sociales o no (social_media_hours > 0)
data['hace_ejercicio'] = data['exercise_minutes'].apply(lambda x: 1 if x > 0 else 0)
data['juega_videojuegos'] = data['gaming_hours'].apply(lambda x: 1 if x > 0 else 0)
data['usa_redes_sociales'] = data['social_media_hours'].apply(lambda x: 1 if x > 0 else 0)
data
```

```
[15]: student_id  age  gender  academic_level  study_hours  self_study_hours  \
0           1  20.0     0.0                1         5.37           2.09
1           2  16.0     1.0                0         5.85           5.04
2           3  18.0     1.0                1         5.69           2.27
```

|      |      |      |     |     |      |      |
|------|------|------|-----|-----|------|------|
| 3    | 4    | 24.0 | 0.0 | 1   | 2.32 | 1.06 |
| 4    | 5    | 24.0 | 1.0 | 2   | 3.87 | 2.63 |
| ...  | ...  | ...  | ... | ... | ...  | ...  |
| 5612 | 5613 | 18.0 | 0.0 | 2   | 4.30 | 2.34 |
| 5614 | 5615 | 25.0 | 1.0 | 2   | 3.85 | 2.63 |
| 5615 | 5616 | 17.0 | 0.0 | 0   | 6.55 | 2.48 |
| 5616 | 5617 | 21.0 | 0.0 | 2   | 4.16 | 0.00 |
| 5620 | 5621 | 18.0 | 0.0 | 2   | 3.96 | 0.98 |

|      | online_classes_hours | social_media_hours | gaming_hours | sleep_hours | \ |
|------|----------------------|--------------------|--------------|-------------|---|
| 0    | 1.85                 | 3.66               | 2.32         | 7.73        |   |
| 1    | 1.87                 | 3.60               | 2.79         | 6.11        |   |
| 2    | 0.00                 | 2.93               | 3.94         | 7.11        |   |
| 3    | 2.30                 | 4.34               | 2.37         | 8.54        |   |
| 4    | 2.54                 | 3.71               | 1.30         | 7.69        |   |
| ...  | ...                  | ...                | ...          | ...         |   |
| 5612 | 0.70                 | 0.43               | 4.04         | 7.39        |   |
| 5614 | 2.01                 | 2.55               | 2.46         | 8.15        |   |
| 5615 | 2.01                 | 3.42               | 2.44         | 6.17        |   |
| 5616 | 1.27                 | 4.01               | 3.41         | 8.07        |   |
| 5620 | 1.77                 | 4.35               | 2.22         | 6.57        |   |

|      | internet_quality | mental_health_score | focus_index | burnout_level | \ |
|------|------------------|---------------------|-------------|---------------|---|
| 0    | 2                | 3.0                 | 19.01       | 31.77         |   |
| 1    | 2                | 10.0                | 42.10       | 45.89         |   |
| 2    | 1                | 5.0                 | 21.93       | 37.07         |   |
| 3    | 0                | 3.0                 | 13.47       | 43.63         |   |
| 4    | 0                | 2.0                 | 19.95       | 56.62         |   |
| ...  | ...              | ...                 | ...         | ...           |   |
| 5612 | 0                | 2.0                 | 36.18       | 40.42         |   |
| 5614 | 2                | 10.0                | 38.26       | 57.39         |   |
| 5615 | 0                | 7.0                 | 34.70       | 56.14         |   |
| 5616 | 0                | 7.0                 | 21.78       | 48.07         |   |
| 5620 | 0                | 3.0                 | 16.28       | 36.62         |   |

|      | productivity_score | exam_score | rinde | hace_ejercicio | \ |
|------|--------------------|------------|-------|----------------|---|
| 0    | 42.59              | 25.080000  | 1     | 1              |   |
| 1    | 67.15              | 37.830000  | 1     | 1              |   |
| 2    | 37.68              | 18.660000  | 1     | 1              |   |
| 3    | 12.83              | 1.000000   | 0     | 1              |   |
| 4    | 18.53              | 7.780000   | 1     | 1              |   |
| ...  | ...                | ...        | ...   | ...            |   |
| 5612 | 31.42              | 10.570000  | 1     | 1              |   |
| 5614 | 45.30              | 23.770000  | 1     | 1              |   |
| 5615 | 47.66              | 20.616782  | 1     | 1              |   |
| 5616 | 43.06              | 17.150000  | 1     | 1              |   |
| 5620 | 21.36              | 2.810000   | 1     | 1              |   |

|      | juega_videojuegos | usa_redes_sociales |
|------|-------------------|--------------------|
| 0    | 1                 | 1                  |
| 1    | 1                 | 1                  |
| 2    | 1                 | 1                  |
| 3    | 1                 | 1                  |
| 4    | 1                 | 1                  |
| ...  | ...               | ...                |
| 5612 | 1                 | 1                  |
| 5614 | 1                 | 1                  |
| 5615 | 1                 | 1                  |
| 5616 | 1                 | 1                  |
| 5620 | 1                 | 1                  |

[4994 rows x 25 columns]

Creamos un mapa de calor con las correlaciones y así ver las variables más importantes para nuestro modelo

```
[16]: # Crear una matriz de correlación para data
correlation_matrix = data.corr()

# Visualizar la matriz de correlación con un heatmap
plt.figure(figsize=(14, 10))
sns.heatmap(correlation_matrix, annot=True, fmt='.2f', cmap='coolwarm',
            center=0, square=True)
plt.title('Matriz de Correlación')
plt.tight_layout()
plt.show()
```



```
print(modelo.summary())
```

```

                                OLS Regression Results
=====
Dep. Variable:                  rinde      R-squared:                0.240
Model:                          OLS      Adj. R-squared:           0.239
Method:                        Least Squares  F-statistic:              117.9
Date:                          Wed, 22 Apr 2026  Prob (F-statistic):    1.14e-139
Time:                          15:09:57    Log-Likelihood:           -70.143
No. Observations:              4994      AIC:                      154.3
Df Residuals:                  4987      BIC:                      199.9
Df Model:                      6
Covariance Type:               HC1
=====
=====
                                coef      std err          z      P>|z|      [0.025
0.975]
-----
Intercept                    0.7013      0.036      19.275      0.000      0.630
0.773
productivity_score           0.0055      0.001       8.214      0.000      0.004
0.007
focus_index                 0.0037      0.001       5.733      0.000      0.002
0.005
mental_health_score          0.0004      0.002       0.186      0.853     -0.004
0.005
study_hours                  0.0024      0.004       0.591      0.555     -0.006
0.010
burnout_level                -0.0025      0.000      -7.259      0.000     -0.003
-0.002
sleep_hours                  -0.0003      0.004      -0.078      0.938     -0.007
0.007
=====
Omnibus:                    1720.077    Durbin-Watson:              2.027
Prob(Omnibus):              0.000    Jarque-Bera (JB):          4873.215
Skew:                       -1.848    Prob(JB):                  0.00
Kurtosis:                   6.123    Cond. No.                  682.
=====

```

Notes:

[1] Standard Errors are heteroscedasticity robust (HC1)

Podemos apreciar que las variables con mayor coeficiente son productivity\_score, focus\_index, study\_hours y burnout\_level, siendo esta última la única, junto con sleep\_hours, con coeficiente negativo.

mental\_health\_score, study\_hours y sleep\_hours resultaron no ser significativas, mientras que el resto si cuentan con un valor  $P < 0.05$ .



El R-squared es bastante bajo, por lo que el modelo no explica en su mayoría nuestra variable dependiente.

### 1.0.3 3. Ejecute un modelo *probit* para responder a la pregunta 2. Seleccione las variables dependientes a incluir en el modelo final e interprete su significado.

```
[18]: probit = smf.probit(formula, data=data).fit(cov_type='HC1')
print(probit.summary())
```

Optimization terminated successfully.

Current function value: 0.129994

Iterations 9

#### Probit Regression Results

```
=====
Dep. Variable:          rinde    No. Observations:          4994
Model:                  Probit    Df Residuals:              4987
Method:                  MLE      Df Model:                  6
Date:                   Wed, 22 Apr 2026    Pseudo R-squ.:          0.5591
Time:                   15:09:57    Log-Likelihood:         -649.19
converged:              True      LL-Null:                -1472.5
Covariance Type:        HC1      LLR p-value:            0.000
=====
```

```
=====
                                coef      std err          z      P>|z|      [0.025
0.975]
-----
Intercept                   -0.2188      0.390      -0.561      0.575      -0.984
0.546
productivity_score           0.0851      0.010       8.740      0.000       0.066
0.104
focus_index                 0.0488      0.007       6.651      0.000       0.034
0.063
mental_health_score          0.0357      0.031       1.151      0.250      -0.025
0.096
study_hours                  0.0186      0.051       0.364      0.716      -0.081
0.119
burnout_level               -0.0347      0.004      -8.497      0.000      -0.043
-0.027
sleep_hours                  0.0069      0.041       0.169      0.866      -0.073
0.087
=====
```

Possibly complete quasi-separation: A fraction 0.39 of observations can be perfectly predicted. This might indicate that there is complete quasi-separation. In this case some parameters will not be identified.

Los resultados, tanto en coeficientes como en significancia, son similares, aunque con la diferencia que en este caso los coeficientes son considerablemente más altos, `burnout_level` ahora es la única variable con coeficiente negativo y se puede apreciar que la salud mental aumentó su significancia, mientras que las horas de estudio disminuyeron la suya.

El pseudo R-squared es mayor en este caso, por lo que si podemos decir que el modelo ya explica lo suficiente nuestra variable (en comparación con el modelo anterior).

#### 1.0.4 4. Ejecute un modelo *logit* para responder a la pregunta 2. Seleccione las variables dependientes a incluir en el modelo final e interprete su significado.

```
[19]: logit = smf.logit(formula, data=data).fit()
      print(logit.summary())

      mfx = logit.get_margeff()
      print(mfx.summary())
```

Optimization terminated successfully.

Current function value: 0.130302

Iterations 9

##### Logit Regression Results

```
=====
Dep. Variable:          rinde    No. Observations:          4994
Model:                  Logit    Df Residuals:              4987
Method:                  MLE     Df Model:                  6
Date:                    Wed, 22 Apr 2026    Pseudo R-squ.:        0.5581
Time:                    15:09:57    Log-Likelihood:       -650.73
converged:                True    LL-Null:              -1472.5
Covariance Type:          nonrobust    LLR p-value:          0.000
=====
```

```
=====
                                coef    std err          z      P>|z|      [0.025
0.975]
-----
Intercept                   -0.5189      0.699      -0.742      0.458     -1.889
0.851
productivity_score           0.1666      0.015     11.157      0.000      0.137
0.196
focus_index                 0.0849      0.012      6.908      0.000      0.061
0.109
mental_health_score          0.0439      0.053      0.836      0.403     -0.059
0.147
study_hours                 -0.0121      0.082     -0.148      0.883     -0.172
0.148
burnout_level               -0.0606      0.007     -8.215      0.000     -0.075
-0.046
sleep_hours                  0.0234      0.072      0.326      0.744     -0.117
```

0.164

=====  
=====

Possibly complete quasi-separation: A fraction 0.19 of observations can be perfectly predicted. This might indicate that there is complete quasi-separation. In this case some parameters will not be identified.

#### Logit Marginal Effects

=====  
Dep. Variable:                      rinde  
Method:                              dydx  
At:                                    overall  
=====

|                     | dy/dx   | std err | z      | P> z  | [0.025 |
|---------------------|---------|---------|--------|-------|--------|
| 0.975]              |         |         |        |       |        |
| -----               |         |         |        |       |        |
| productivity_score  | 0.0065  | 0.001   | 12.175 | 0.000 | 0.005  |
| 0.008               |         |         |        |       |        |
| focus_index         | 0.0033  | 0.000   | 7.130  | 0.000 | 0.002  |
| 0.004               |         |         |        |       |        |
| mental_health_score | 0.0017  | 0.002   | 0.837  | 0.403 | -0.002 |
| 0.006               |         |         |        |       |        |
| study_hours         | -0.0005 | 0.003   | -0.148 | 0.883 | -0.007 |
| 0.006               |         |         |        |       |        |
| burnout_level       | -0.0024 | 0.000   | -8.594 | 0.000 | -0.003 |
| -0.002              |         |         |        |       |        |
| sleep_hours         | 0.0009  | 0.003   | 0.326  | 0.744 | -0.005 |
| 0.006               |         |         |        |       |        |
| =====               |         |         |        |       |        |
| =====               |         |         |        |       |        |

Resultados bien similares al del caso anterior, con coeficientes aún mayores, las horas de estudio ahora afectan negativamente y las significancias también cambiaron de forma similar que en el caso anterior.

**1.0.5 5. Comente los resultados obtenidos en 2, 3 y 4. ¿Cuáles y por qué existen las diferencias entre los resultados?. En su opinión, ¿Cuál sería el más adecuado para responder la pregunta de investigación y por qué? ¿Qué variables resultaron ser robustas a la especificación?**

```
[21]: stargazer = Stargazer([modelo, probit, logit])

# 3. Configuración de la visualización
stargazer.title("Comparación de Modelos: MCO, Probit y Logit")
stargazer.custom_columns(["Modelo MCO", "Modelo Probit", "Modelo Logit"], [1, 1, 1])
```

```
stargazer.add_custom_notes(["Nota: Todos los modelos utilizan errores estándar_
↳robustos (HC1)."])
```

```
# 4. Renderizado para Jupyter Notebook
```

```
HTML(stargazer.render_html())
```

[21]: <IPython.core.display.HTML object>

Debido a la naturaleza de los modelos, Probit y Logit son los mejores para este tipo de análisis.

### 1.0.6 6. Use un modelo Poisson para explicar la nota del examen, entre aquellos alumnos que lo rindieron. Seleccione las variables dependientes a incluir en el modelo final e interprete su significado.

Filtramos los datos para considerar solo quienes rindieron el examen

```
[ ]: data_examen = data[data['rinde'] == 1]
```

Para el modelo agregaré además la variable 'upcoming\_deadline', 'part\_time\_job' y 'screen\_time\_hours'

```
[27]: formula_ps = '''exam_score ~ productivity_score + focus_index +_
↳mental_health_score + study_hours + burnout_level + sleep_hours +_
↳burnout_level + upcoming_deadline + part_time_job + screen_time_hours'''
modelo_ps = smf.poisson(formula_ps, data=data_examen).fit(cov_type='HC1')
print(modelo_ps.summary())
```

Optimization terminated successfully.

Current function value: 3.290639

Iterations 5

#### Poisson Regression Results

```
=====
Dep. Variable:          exam_score  No. Observations:          4561
Model:                  Poisson    Df Residuals:              4551
Method:                  MLE       Df Model:                  9
Date:                   mi., 22 abr. 2026  Pseudo R-squ.:          0.3995
Time:                   15:30:07    Log-Likelihood:         -15009.
converged:               True      LL-Null:                 -24993.
Covariance Type:        HC1      LLR p-value:             0.000
=====
=====
                                coef      std err          z      P>|z|      [0.025
0.975]
-----
Intercept                2.0175      0.049      41.185      0.000      1.921
2.113
productivity_score        0.0186      0.001     21.394      0.000      0.017
0.020
```

|                     |         |       |         |       |        |
|---------------------|---------|-------|---------|-------|--------|
| focus_index         | 0.0126  | 0.001 | 17.133  | 0.000 | 0.011  |
| 0.014               |         |       |         |       |        |
| mental_health_score | 0.0144  | 0.003 | 4.414   | 0.000 | 0.008  |
| 0.021               |         |       |         |       |        |
| study_hours         | 0.0127  | 0.005 | 2.403   | 0.016 | 0.002  |
| 0.023               |         |       |         |       |        |
| burnout_level       | -0.0082 | 0.001 | -13.242 | 0.000 | -0.009 |
| -0.007              |         |       |         |       |        |
| sleep_hours         | 0.0054  | 0.005 | 1.089   | 0.276 | -0.004 |
| 0.015               |         |       |         |       |        |
| upcoming_deadline   | -0.0330 | 0.011 | -2.930  | 0.003 | -0.055 |
| -0.011              |         |       |         |       |        |
| part_time_job       | -0.0136 | 0.010 | -1.346  | 0.178 | -0.033 |
| 0.006               |         |       |         |       |        |
| screen_time_hours   | -0.0018 | 0.002 | -0.943  | 0.346 | -0.006 |
| 0.002               |         |       |         |       |        |

=====

=====

De las tres variables agregadas, solo upcoming\_deadline fue significativa, pero las tres afectan negativamente a la nota del examen.

### 1.0.7 7. Determine si existe sobre dispersion en la data y posible valor optimo de alpha para un modelo Binomial Negativa.

```
[28]: # Se compara media y varianza
media = data_examen['exam_score'].mean()
varianza = data_examen['exam_score'].var()

print(f"Media: {media}")
print(f"Varianza: {varianza}")
```

Media: 20.61678244972578  
 Varianza: 122.18590269668688

Como la varianza es muy mayor a la media, podemos suponer que existe una sobre dispersión en los datos

```
[31]: # Test de Overdispersion.
aux=((data_examen['exam_score']-modelo_ps.predict())**2-modelo_ps.predict())/
    modelo_ps.predict()
auxr=sm.OLS(aux,modelo_ps.predict()).fit()
print(auxr.summary())
```

#### OLS Regression Results

=====

=====

Dep. Variable: exam\_score R-squared (uncentered): 0.035

```

Model:                                OLS    Adj. R-squared (uncentered):
0.035
Method:                             Least Squares    F-statistic:
165.3
Date:                               mi., 22 abr. 2026    Prob (F-statistic):
3.40e-37
Time:                               15:44:48    Log-Likelihood:
-11351.
No. Observations:                   4561    AIC:
2.270e+04
Df Residuals:                       4560    BIC:
2.271e+04
Df Model:                           1
Covariance Type:                    nonrobust
=====
              coef      std err          t      P>|t|      [0.025      0.975]
-----
x1              0.0242      0.002     12.858      0.000      0.020      0.028
=====
Omnibus:                        4591.916    Durbin-Watson:                1.994
Prob(Omnibus):                   0.000    Jarque-Bera (JB):            379080.436
Skew:                            4.805    Prob(JB):                     0.00
Kurtosis:                       46.616    Cond. No.                     1.00
=====

```

#### Notes:

[1]  $R^2$  is computed without centering (uncentered) since the model does not contain a constant.

[2] Standard Errors assume that the covariance matrix of the errors is correctly specified.

Se realizó un test de sobredispersión, donde se puede concluir que efectivamente existe una sobredispersión. Además podemos obtener el  $\alpha$  óptimo para un modelo de binomial negativa, siendo este = 0.0242.

**1.0.8 8. Usando la informacion anterior, ejecute un modelo Binomial Negativa para responder a la pregunta 6. Seleccione las variables dependientes a incluir en el modelo final e interprete su significado.**

```

[32]: # Para calcular alpha:
modelo_nb = smf.negativebinomial(formula_ps, data=data_examen).fit()

print(modelo_nb.summary())

```

Optimization terminated successfully.

Current function value: 3.218068

Iterations: 27

Function evaluations: 46

Gradient evaluations: 39

### NegativeBinomial Regression Results

```
=====
Dep. Variable:          exam_score    No. Observations:          4561
Model:                 NegativeBinomial    Df Residuals:              4551
Method:                MLE              Df Model:                  9
Date:                  mi., 22 abr. 2026    Pseudo R-squ.:            0.1540
Time:                  15:46:27            Log-Likelihood:           -14678.
converged:              True              LL-Null:                  -17350.
Covariance Type:        nonrobust          LLR p-value:              0.000
=====
```

```
=====
                                coef      std err          z      P>|z|      [0.025
0.975]
```

```
-----
Intercept                1.9625        0.051     38.814     0.000        1.863
2.062
productivity_score        0.0195        0.001     25.034     0.000        0.018
0.021
focus_index              0.0138        0.001     18.280     0.000        0.012
0.015
mental_health_score       0.0139        0.003      4.585     0.000        0.008
0.020
study_hours               0.0141        0.005      2.923     0.003        0.005
0.024
burnout_level             -0.0086        0.001    -13.981     0.000       -0.010
-0.007
sleep_hours               0.0060        0.005      1.140     0.254       -0.004
0.016
upcoming_deadline        -0.0380        0.012     -3.189     0.001       -0.061
-0.015
part_time_job             -0.0171        0.011     -1.613     0.107       -0.038
0.004
screen_time_hours        -0.0028        0.002     -1.310     0.190       -0.007
0.001
alpha                     0.0368        0.002     17.253     0.000        0.033
0.041
=====
```

Podemos apreciar que, aunque siguen habiendo variables poco significativas, estas aumentaron su significancia.

1.0.9 9. Comente los resultados obtenidos en 6, 7 y 8. ¿Cuáles y por qué existen las diferencias entre los resultados?. En su opinión, ¿Cuál sería el más adecuado para responder la pregunta de investigación y por qué? ¿Qué variables resultaron ser robustas a la especificación?

```
[33]: stargazer_conteo = Stargazer([modelo_ps, modelo_nb])
stargazer_conteo.title('Comparación de Modelos: Poisson vs Binomial Negativo')
stargazer_conteo.add_custom_notes([
    "Errores estándar robustos HC1 entre paréntesis.",
    'Alpha en BN confirma sobredispersión.'])

HTML(stargazer_conteo.render_html())
```

[33]: <IPython.core.display.HTML object>

Ambos modelos son bastante similares en cuanto a coeficientes y variables robustas.

Variables robustas: - productivity\_score - focus\_index - mental\_health\_score - study\_hours - burnout\_level - upcoming\_deadline

Variables no robustas: - sleep\_hours - part\_time\_job - screen\_time\_hours

La principal diferencia entre ambos modelos radica en que el modelo poisson presenta sobredispersión, la cual es corregida mediante la binomial negativa y por tanto este modelo es el mejor para este caso.